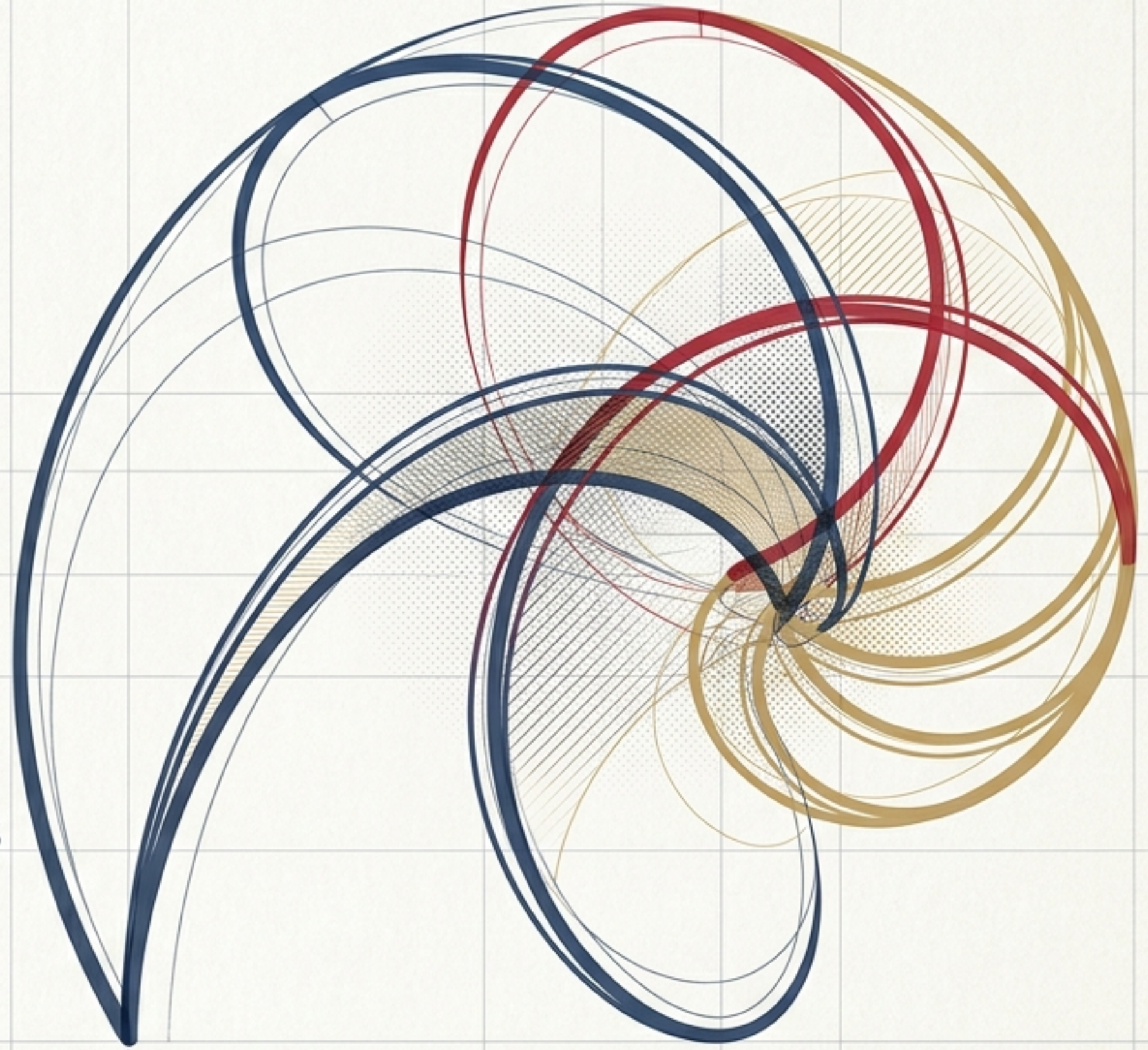
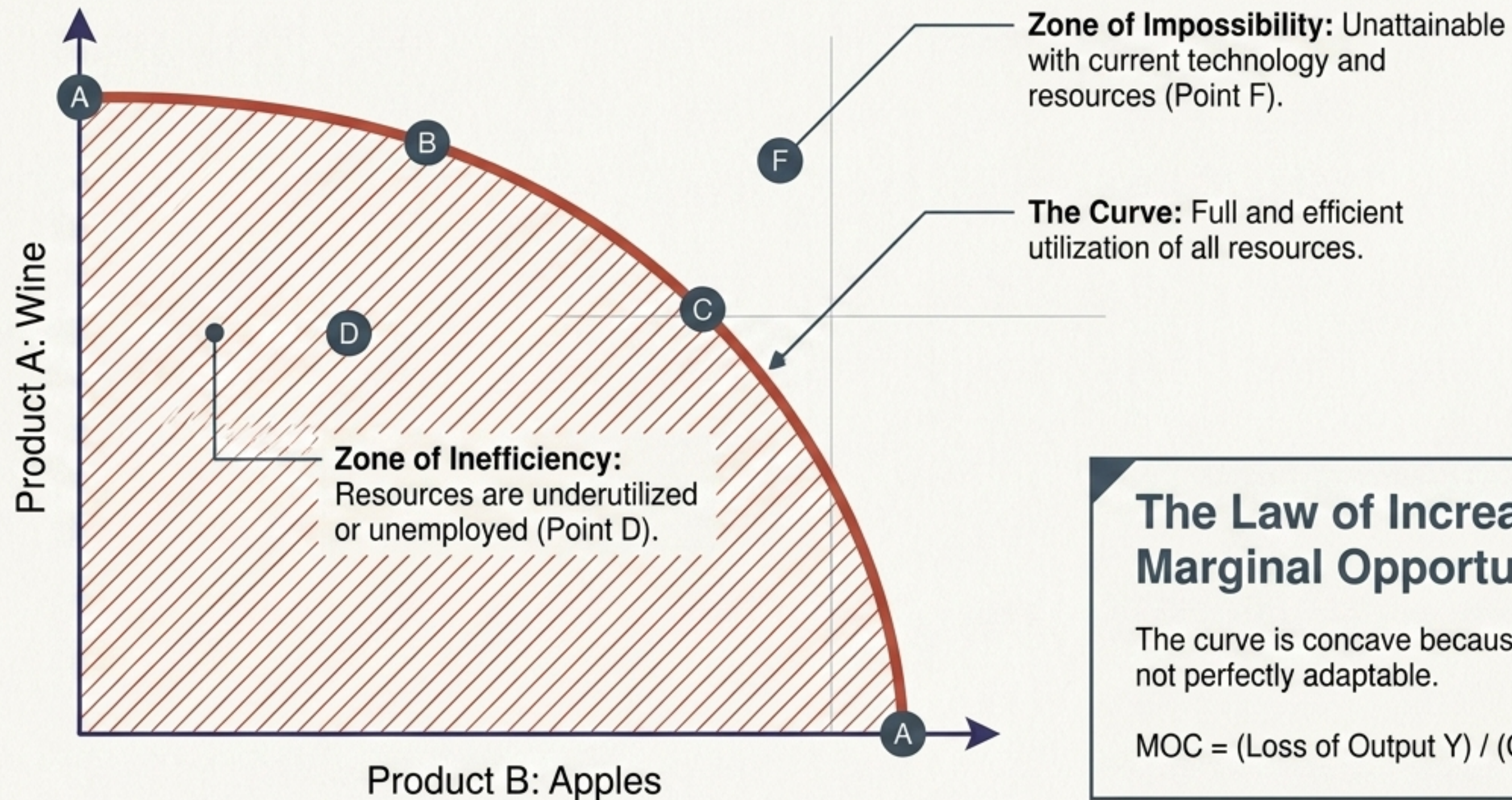


The Architecture of Economic Decisions

Visualizing the Forces of Markets, Production, and Costs



The Constraints of Reality

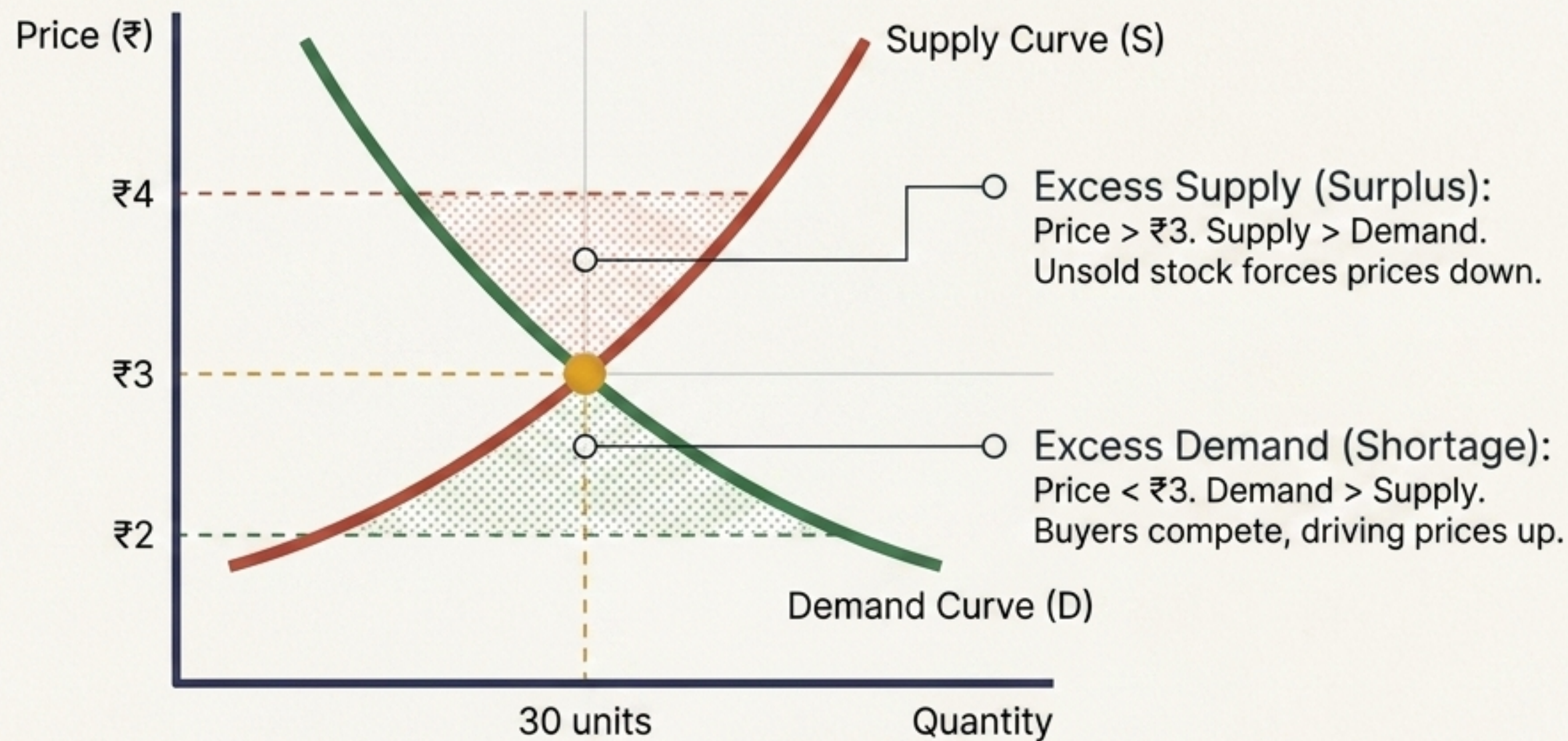


The Law of Increasing Marginal Opportunity Cost

The curve is concave because resources are not perfectly adaptable.

$$\text{MOC} = (\text{Loss of Output Y}) / (\text{Gain of Output X}).$$

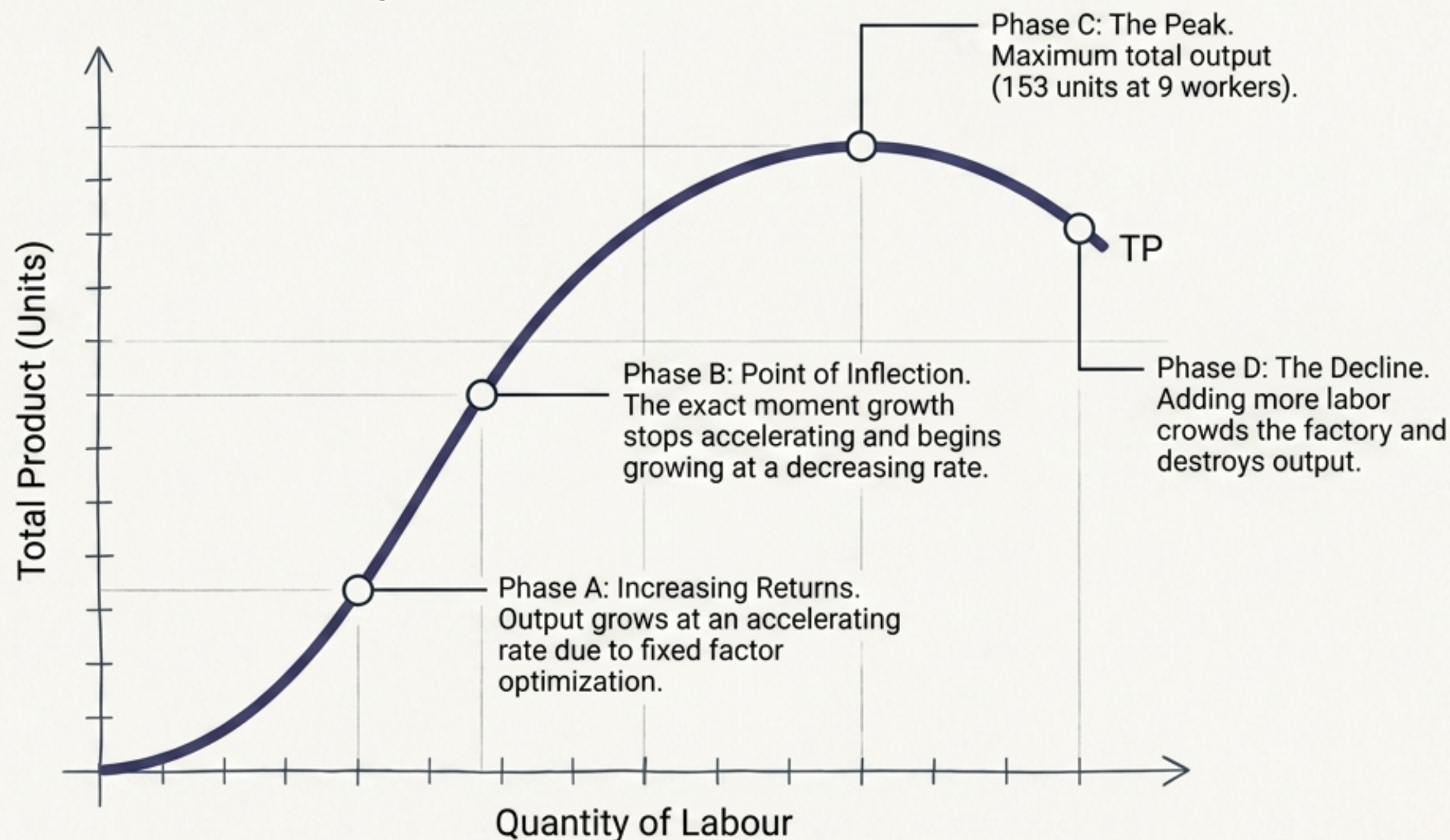
Market Equilibrium



Core Insight: The market is a self-correcting mechanism seeking the exact point where there is no shortage and no excess.

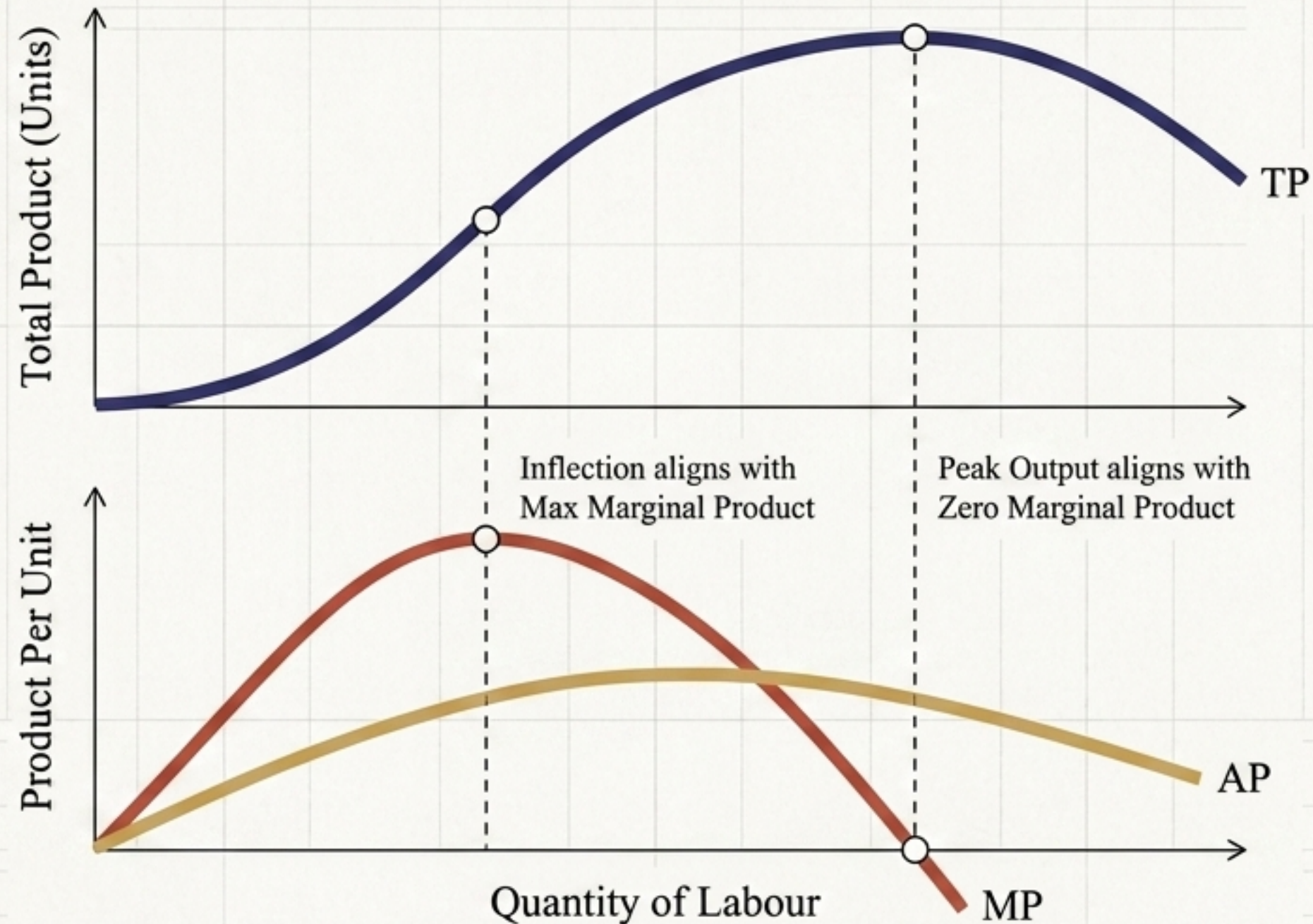
The Law of Variable Proportions

Visualizing the short-run factory floor.



Marginal & Average Dynamics

Looking under the hood of production efficiency.



The Three Stages of Production



Stage I (Increasing Returns)

Geometry:
TP increases at an increasing rate.
MP rises and peaks.

Reality:
Underutilized machinery is finally
being staffed properly.

Stage II (Diminishing Returns)

Geometry:
TP increases at a decreasing rate.
MP falls but remains positive.
AP begins to fall.

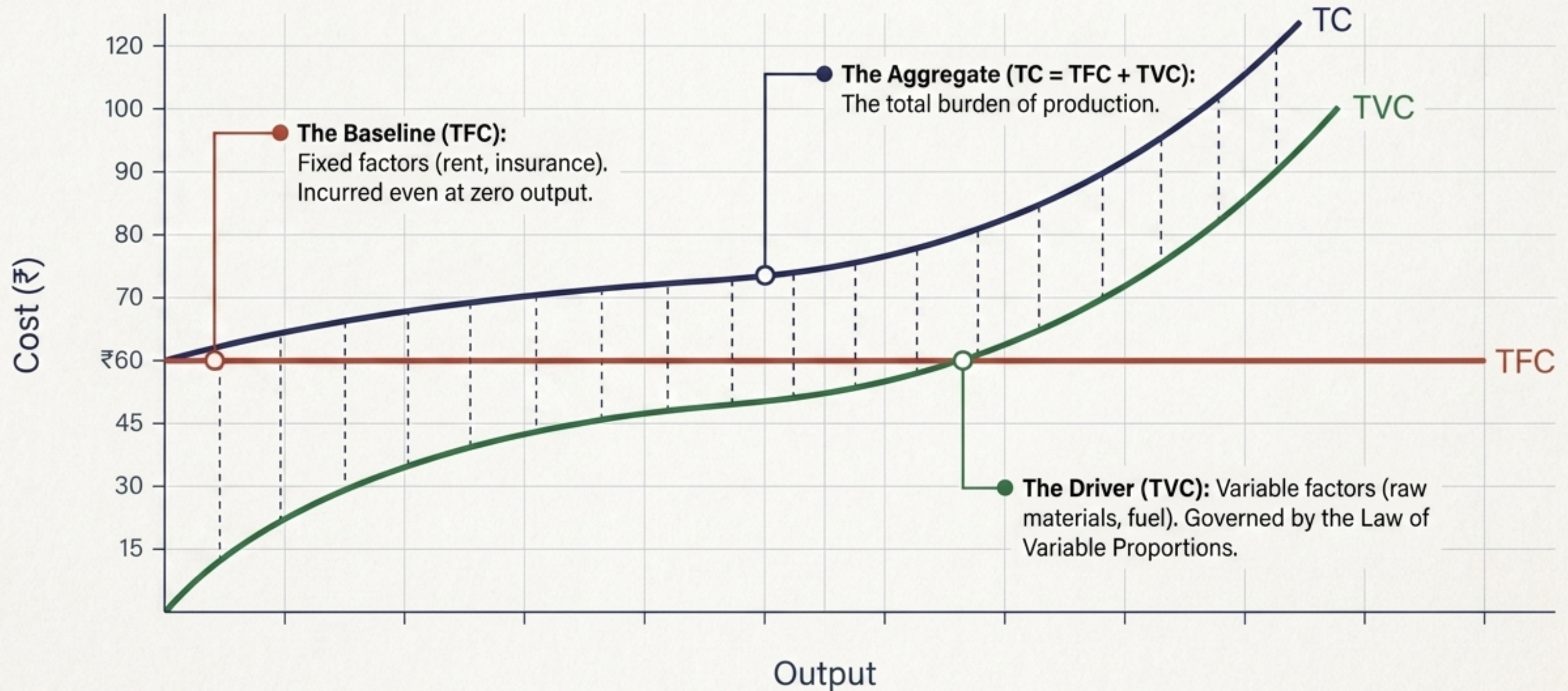
Reality:
Ideal operating zone. The factory
is busy but functional.
(Ends exactly when $MP = 0$).

Stage III (Negative Returns)

Geometry:
TP declines.
MP becomes negative (falls below
zero).

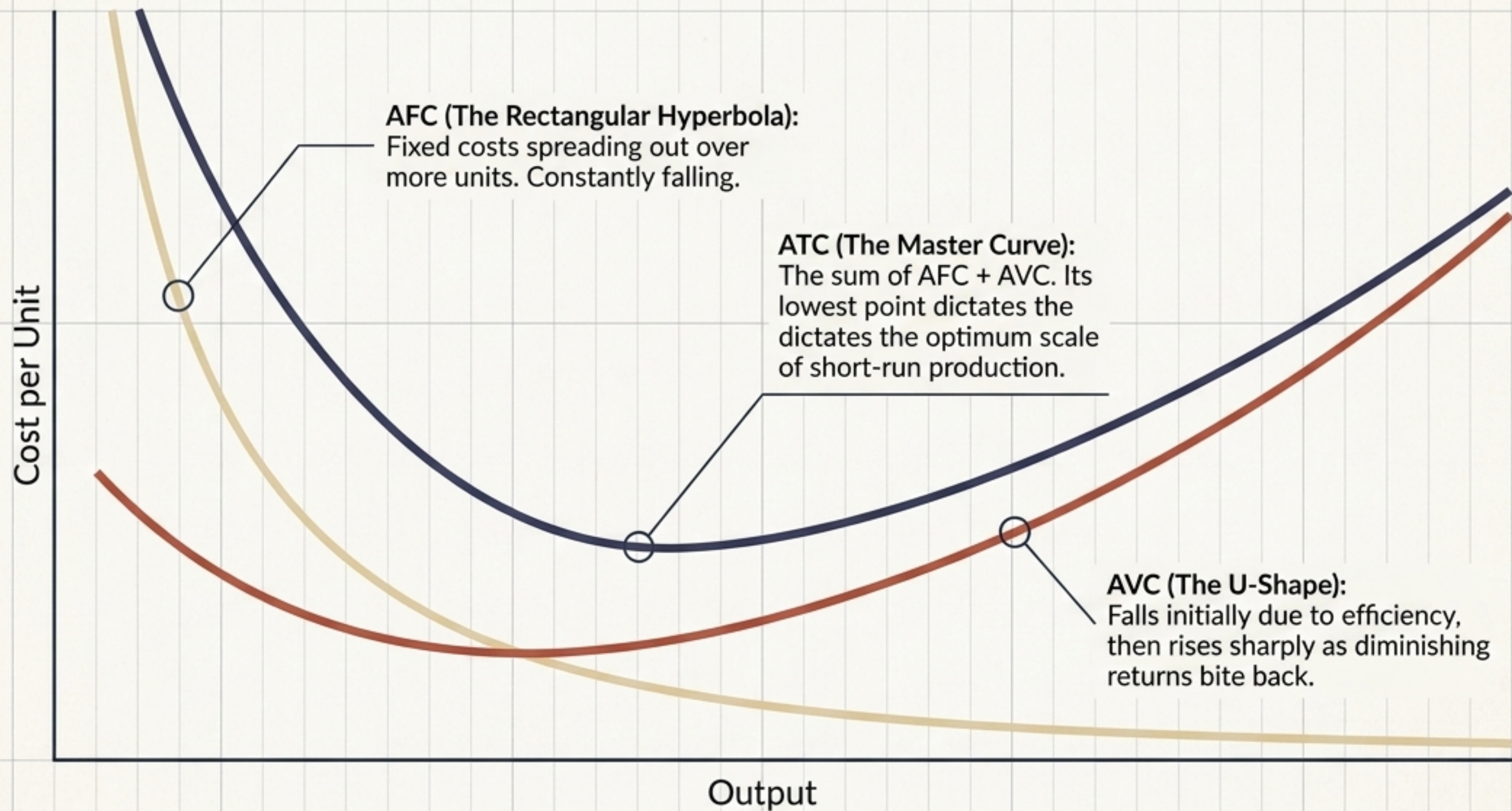
Reality:
Overcrowding. Workers are in
each other's way; adding staff
reduces total output.

The Architecture of Costs: Short-Run Totals

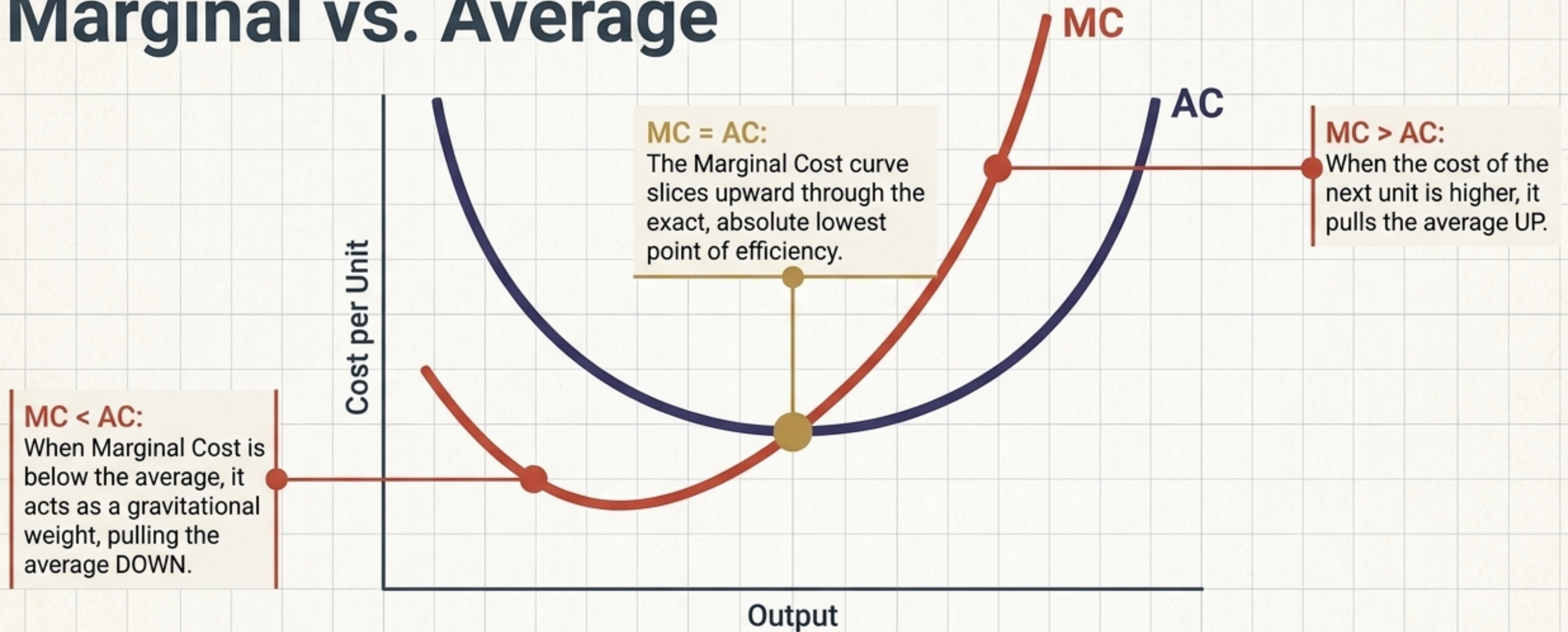


The U-Curve Family

Moving from totals to per-unit realities.



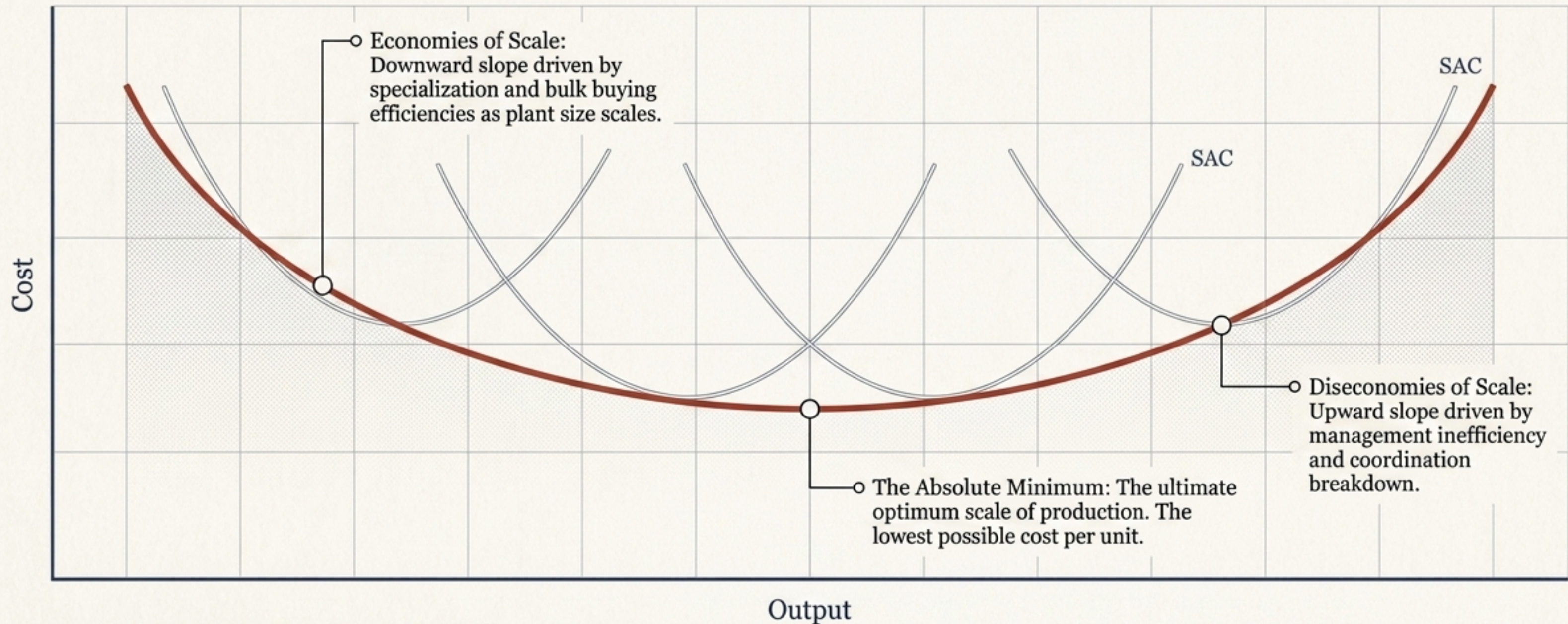
The Golden Rule: Marginal vs. Average



The cost of the **NEXT** unit dictates the **trajectory** of the **AVERAGE** unit.
Therefore, they only meet at the absolute floor of efficiency.

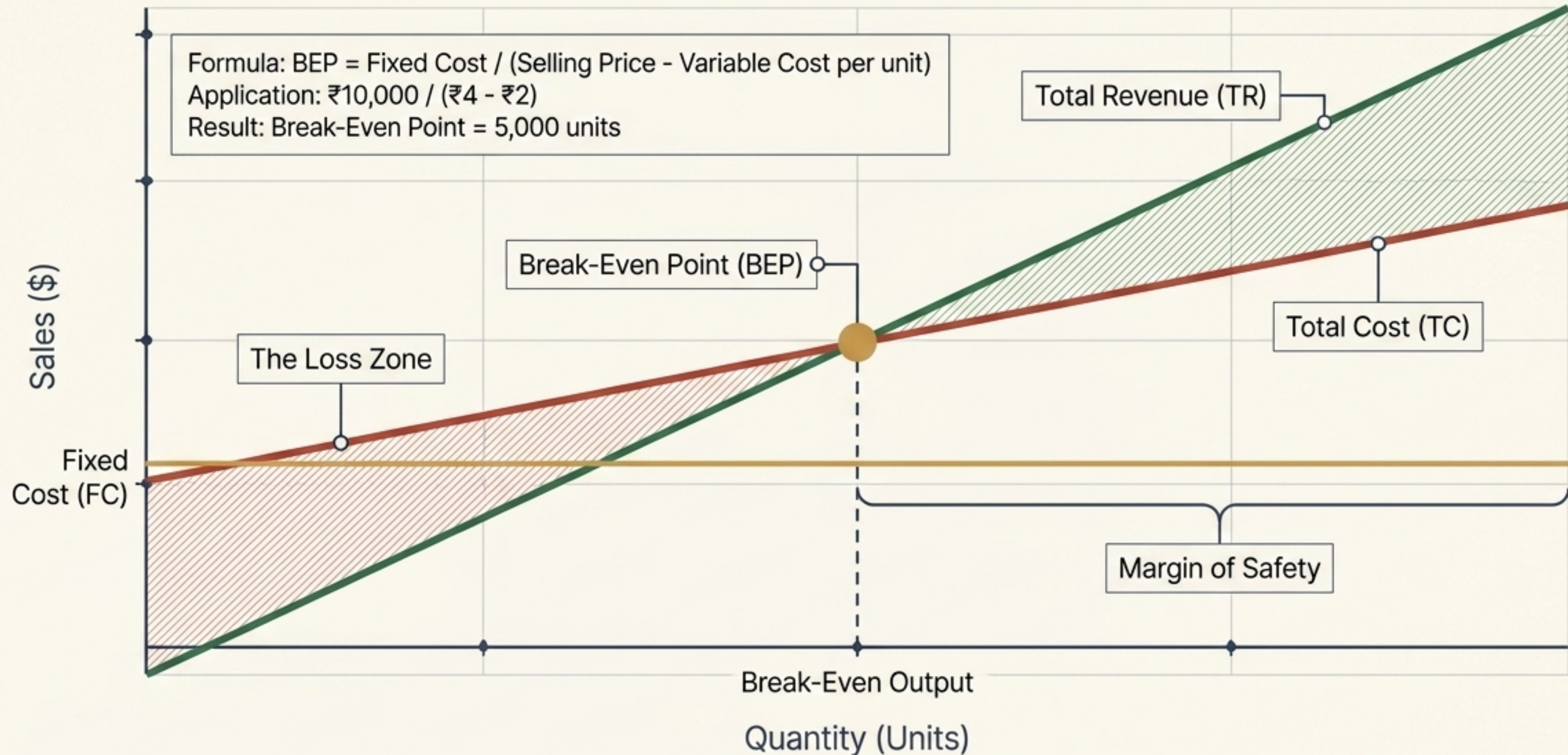
Long-Run Average Cost (LAC)

The Planning Envelope.



The LAC represents the ultimate Planning Curve. In the long run, all factors are variable; the firm can build any plant size.

Survival & The Break-Even Chart



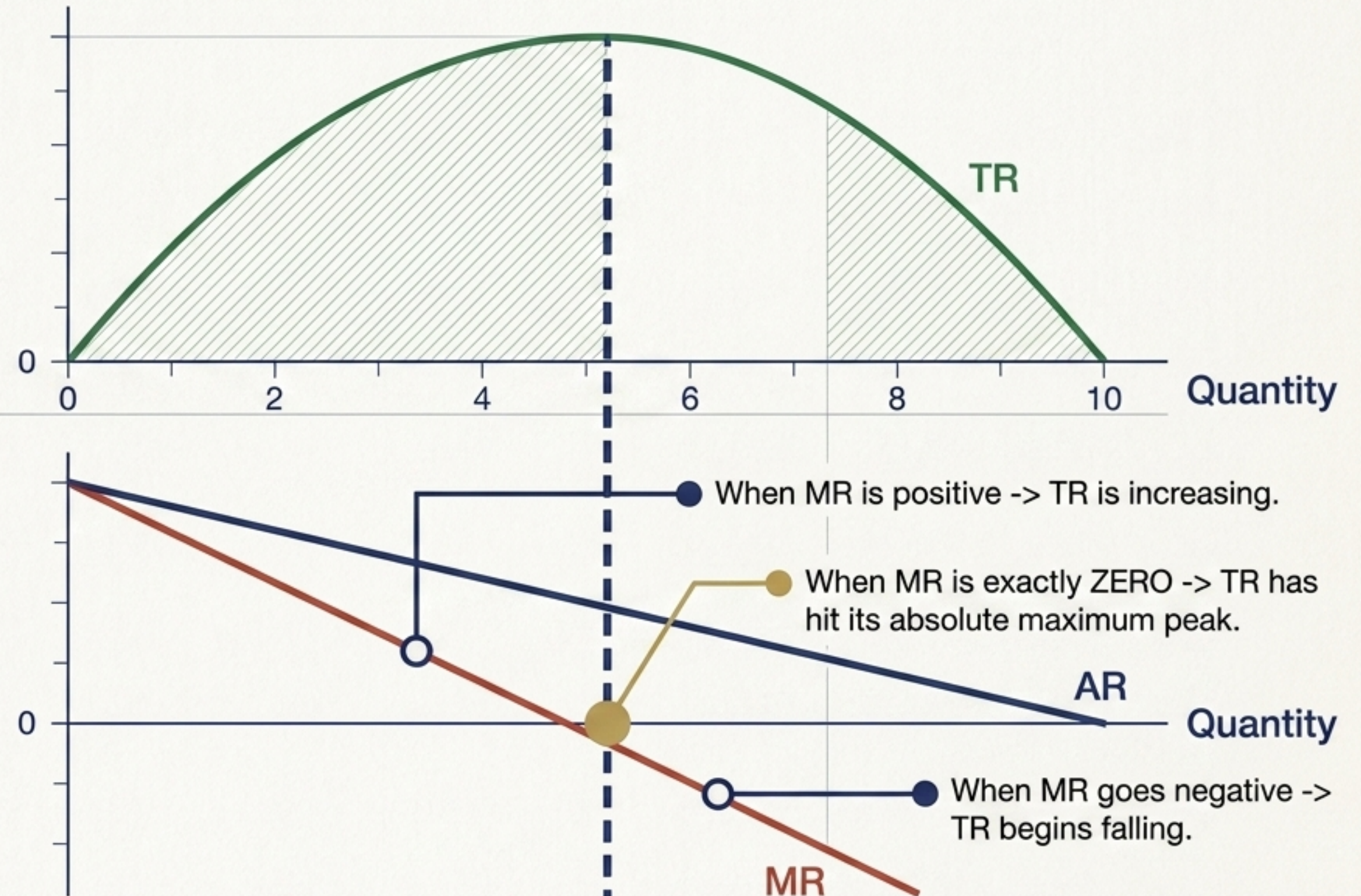
The Geometry of Revenue

$$AR = \frac{TR}{Q}$$

Since $TR = P \times Q$,

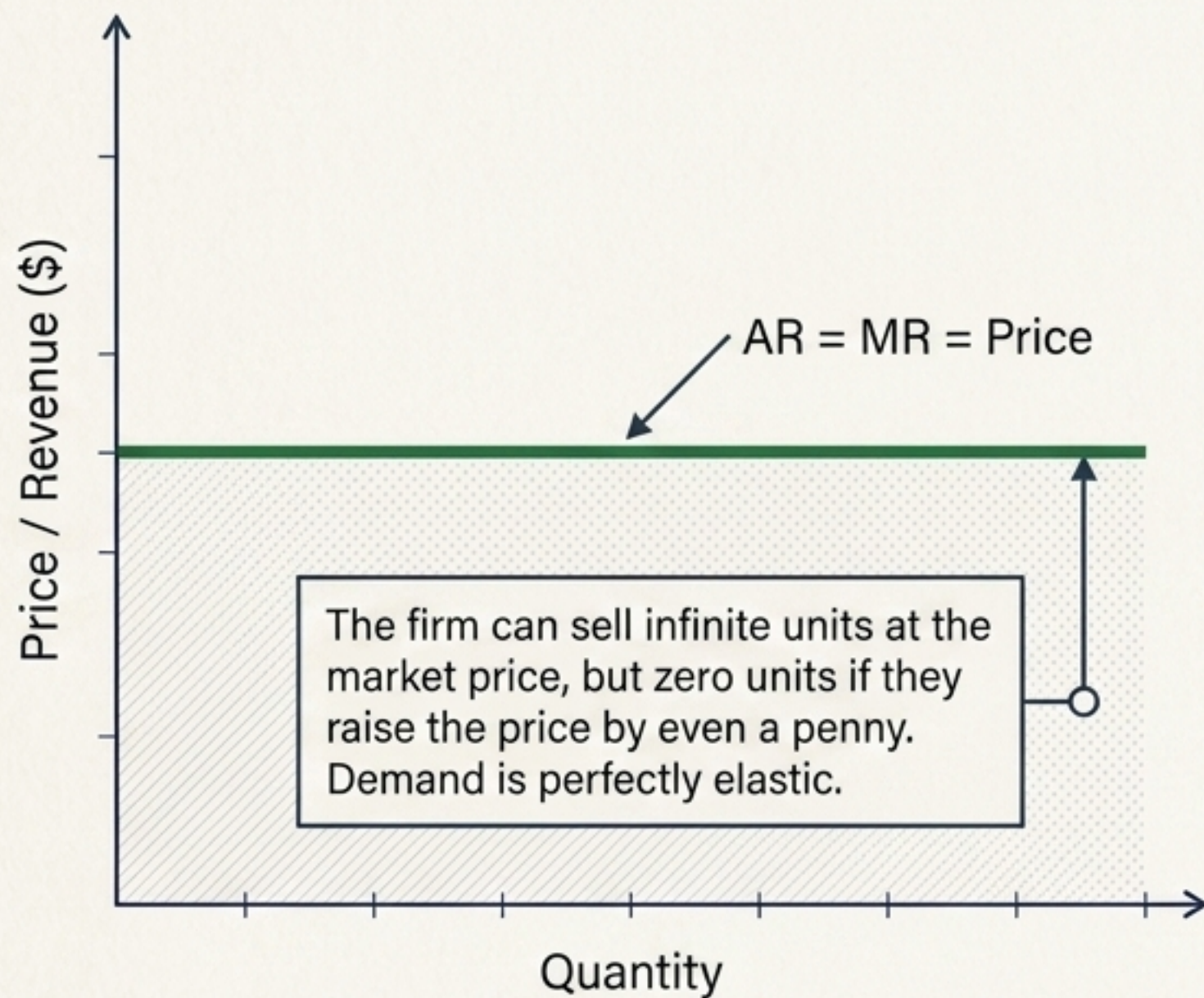
$$AR = \frac{P \times Q}{Q} = P$$

Therefore, Average Revenue is always the Price per unit.

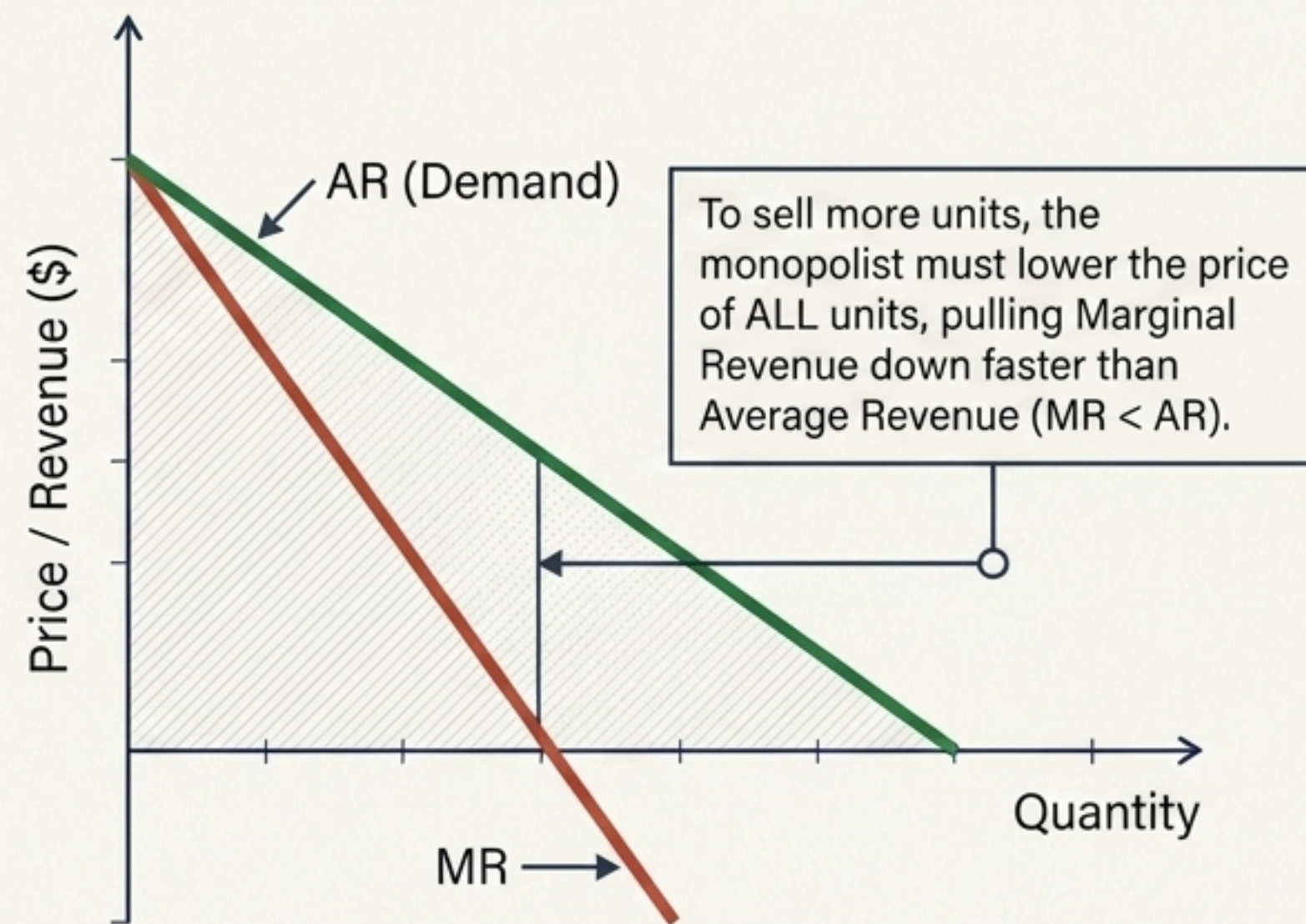


The Market Structure Spectrum

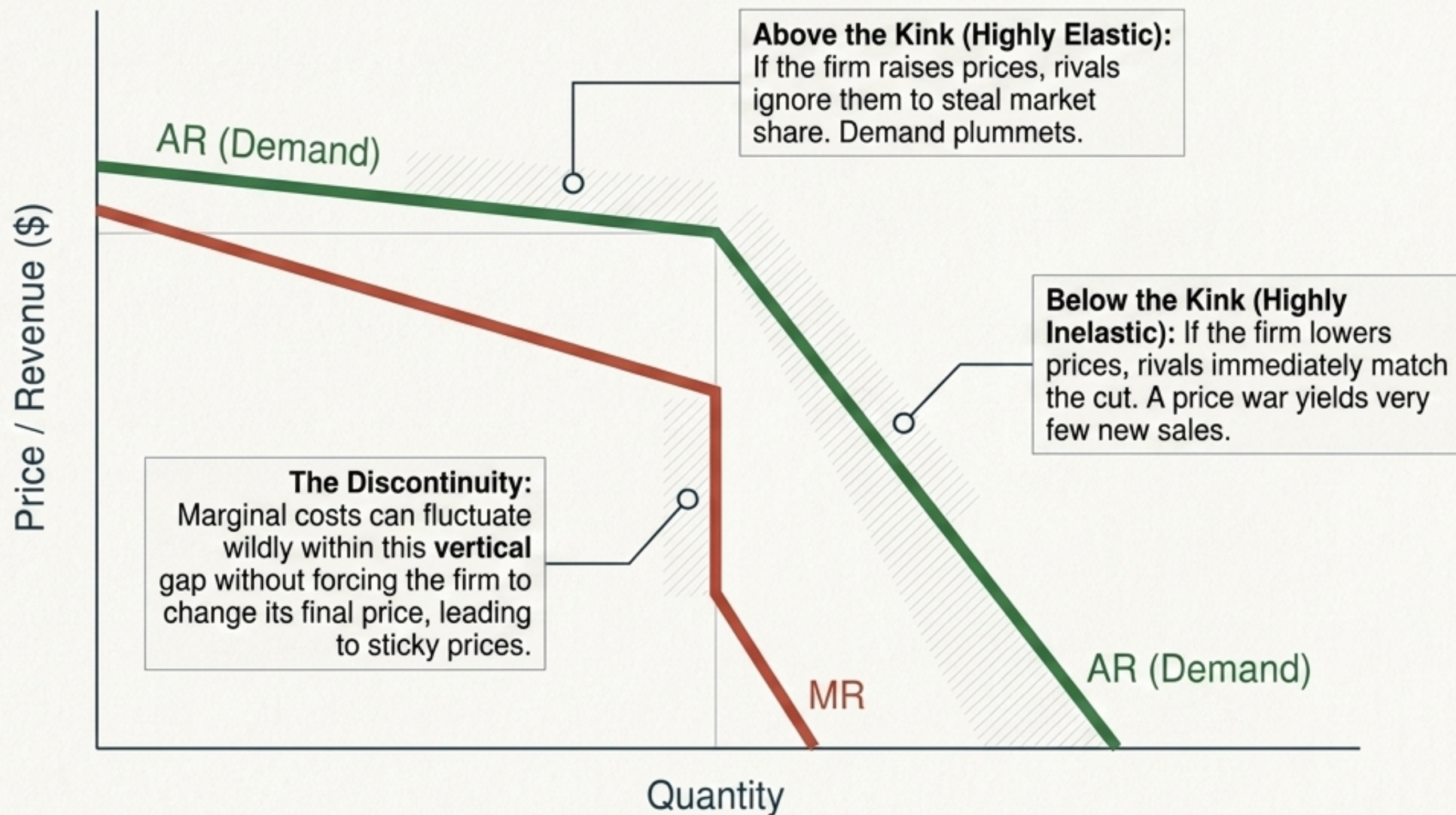
Perfect Competition (Price Taker)



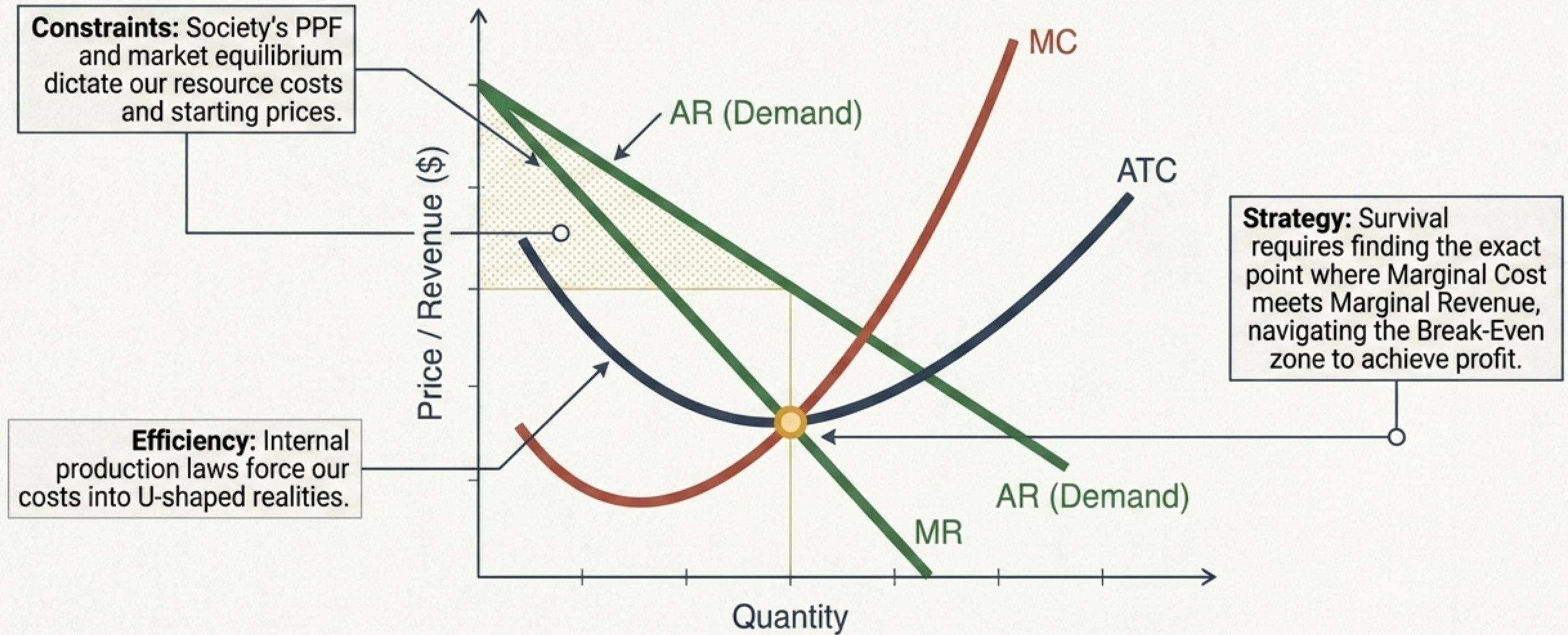
Monopoly (Price Maker)



The Kinked Demand Curve: Oligopoly



The Integrated Firm Blueprint



Economics is not just the study of scarcity; it is the architectural blueprint of human decision-making and value creation.